

Compact dirty controls and a conditional integer-multiplication saving

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Abstract

We replace the full-slot permutations in the simultaneous-layer routine by swaps of compact dirty-control fields. Complete temporary ranges are reserved from existing address coordinates and survive every recursive row split. The row-addition overhead becomes $O(V((f \log p)^\tau + 1))$, with no spacing factor K^τ , and deterministic exceptional repair is included. Together with an independently sized original complex network at $h = 25$, the retained paired bit network and downstream refinements give the conditional witness $\kappa = 83/10^{12} > 2^{-34}$. The underlying manuscript's algorithmic and analytic interfaces remain assumptions. This is a research draft awaiting independent review.

1 Scope and dependencies

The source is OpenAI's *Integer multiplication below $n \log n$* , pinned at

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

The computational model is a fixed finite alphabet and a fixed number of one-dimensional tapes. We retain the published paired-bit construction and its arbitrary-width swap exponent $\tau = 1 - 296/10^{11}$, the nonadjacent CRT/axis routing, the Gaussian width $\alpha = \lceil (32db)^{1/4} \rceil$, the prime selection, resampling, synthetic-transform, error and final rounding interfaces.

The compact-control idea was supplied by an independent research agent. This note supplies the wider-control streaming argument, explicit reservations, repair accounting and recurrence integration needed beyond its address algebra. All temporary values may be arbitrary. No independent address coordinates are added to create scratch space. The exact tests check finite implementations and arithmetic; the general claims rest on the written arguments, not on simulation or formal verification.

2 The replacement movement primitive

2.1 Compact dirty controls for selected-bit additions

The following replacement for the wide-slot packed operation moves only compact control fields. Temporary address fields have arbitrary contents; they are complete ranges of existing address bits, not newly allocated independent coordinates. Their availability in every recursive call is proved separately below.

*Prepared with assistance from OpenAI Codex. This is not independent review or endorsement by OpenAI.

Lemma 1 (Controlled rotations with record-paid offset work). *Consider M records of R bits on a complete rectangular address set, of volume $V = MR$. Its fixed number of field lengths and controls have at most $A = \lceil \log_2(2V) \rceil$ bits each. A binary target field follows all its controls. Suppose its rotation offset, independent of that target, is computable from the controls and descriptors in $O(A^C)$ tape steps for some fixed C . Then the rotation can be performed on fixed tapes in $O(V + MA^{C'})$ time for a fixed C' . In particular, this is $O(V)$ when $A = O(p)$ and R exceeds every fixed power of p .*

Proof. Fix the prefix before the target. If its range is Q and each target block has B suffix bits, rotate by a through the two-piece split at $Q - a$ and merge in the opposite piece order. This is the controlled-rotation construction from the ordered-affine streaming lemma. Each data bit is copied a bounded number of times. There are at most M prefix fibers, since every fiber contains at least one complete record. At each fiber, compute the offset, initialize its two piece lengths, and reset the finite collection of counters; recompute the offset at the merge. Ordinary binary arithmetic and descriptor copies cost a fixed polynomial in A per fiber. The controls can be wider than the target: their lengths are bounded by A , not by the target width. Local suffix-length copies and ripple counters are charged as in the original streaming proof, so no growing descriptor is scanned per payload bit. Two piece tapes, output and fixed arithmetic work tapes suffice; rewinds and clearing are over the regions just written. This gives the stated bound. Finally $MA^{C'}/V = A^{C'}/R = o(1)$ in the specified record regime. $\square$

Let x, y be disjoint slots of fK bits, with corresponding selected positions $j_i = \rho + iK$, $0 \leq i < f$, $0 \leq \rho < K$. Put $n = f - 1$. For $f = 1$ use the elementary two-bit XOR. Otherwise fix a positive integer G , put $B = 2^G$, and assume two disjoint complete nG -bit fields u, t precede both slots and a third such field b follows both slots. Arbitrary spectator fields may intervene. Each compact field has n radix- B digits. The selected-bit operation will preserve these three entire fields.

Earlier source. Suppose x precedes y . The unused field u is a spectator. For an integer target segment v , a bit z , and an arbitrary integer temporary w , perform

$$v \leftarrow v + 2zw, \quad w \leftarrow w + (v \bmod 2), \quad v \leftarrow v + z(1 - 2w), \quad w \leftarrow w - ((v \bmod 2) \oplus z).$$

Writing the original parity as a , the first update leaves it unchanged. The final values are $v + z(1 - 2a)$ and the original w . Thus this toggles only the parity of v when $z = 1$, and restores the dirty temporary.

Apply these four updates simultaneously, with $z_i = x_{j_i}$, to the low n selected segments of y and the digits t_i of t . A target-slot update is one modular rotation of y , whose offsets are respectively

$$\sum_{i < n} 2x_{j_i} t_i 2^{j_i}, \quad \sum_{i < n} x_{j_i} (1 - 2t_i) 2^{j_i} \pmod{2^{fK}}.$$

Both control fields precede y . For a temporary update, swap t with b , rotate the back field by the packed selected bits of the current y , and swap back. For unloading, use minus the packed bits $y_{j_i} \oplus x_{j_i}$. These controls precede the back field. There are four width- nG swaps and four rotations. The arbitrary original contents of b return after each pair of swaps.

Later source. If x follows y , first execute the earlier-source construction with control bits $u_i \bmod 2$. Load the selected bits of x into u by swapping u, b , rotating the back field by their packed radix- B representation, and swapping back. Repeat the earlier-source construction with the new parities of u , then unload the selected bits of x from u by the inverse load. When the digit loads do not

overflow, the two control parities differ by exactly x_{j_i} . The resulting target update is the required XOR, with u, t, b restored. This version uses twelve compact swaps and ten rotations. Every offset depends on a fixed number of fields preceding its target.

Guards and deterministic repair. Temporarily omit the highest selected position. The low n segments $[j_i, j_i + K)$ are disjoint and fit inside y , including when $\rho = K - 1$. Write each initial segment as $2g_i + a_i$, $a_i \in \{0, 1\}$. Declare an address bad if a t digit is $B - 1$, or a target guard fails

$$2B \leq g_i < 2^{K-1} - 2B.$$

For a later source also declare it bad if any u digit is $B - 1$. On a good address each temporary load stays within its digit, and the target displacement is at most $2B$ in either execution of the four-step identity. The conservative guard interval therefore prevents every carry or borrow between the wide segments. After each completed identity the guard is unchanged. Induction through the later-source sequence proves the desired map on the good set, including restoration of every temporary and every unselected address bit.

Uniform complete ranges give the common bound

$$\delta = \min\{1, n(2 \cdot 2^{-G} + 8 \cdot 2^{G-K})\}$$

on the bad fraction; for an earlier source the first factor 2 can be omitted. The ideal map T changes only the selected target parities, so it preserves the bad set. The actual sequence S is a bijection on the entire rectangle: each rotation is independent of its target and all swaps are bijections. Since $S = T$ off the bad set, S preserves that set too. Thus TS^{-1} is supported there, exactly as in the original exceptional-stream repair.

Extract the bad records, retaining marked holes in the full stream. For each extracted address q , reverse the swaps and modular rotations to compute $S^{-1}q$, then compute $T(S^{-1}q)$. All fields and descriptors have $O(A)$ bits. A constant number of offset calculations each scan at most $n \leq A$ digits and combine shifted integers of $O(A)$ bits. Integer arithmetic, mixed-radix decoding, inverse offsets and the destination key can consequently be computed in $O(A^3)$ per record. Sort only the extracted records by their $O(A)$ -bit destination keys, using stable binary radix passes, then reinsert them in their marked holes. Keys are distinct and fill precisely those holes because TS^{-1} permutes the bad set. This uses fixed tapes and costs

$$O(V + MA^3 + \delta MA(R + A)).$$

Every inverse offset is evaluated on the current inverse controls; modular overflow on bad addresses is handled exactly, rather than treated as an integer identity. Finally, supply the omitted highest selected position through the elementary two-bit XOR in $O(V)$ time.

Proposition 1 (Compact-control selected-bit addition). *Under the stated complete-field hypotheses, take*

$$G = 4\lceil \log_2 p \rceil + 6, \quad f \leq p, \quad A = O(p), \quad K/\log p \rightarrow \infty,$$

and let R exceed every fixed power of p . The exact selected-bit row addition, with all temporary fields restored, costs

$$O(V((f \log p)^\tau + 1))$$

on a fixed number of tapes, for either order of the two slots. Constants and cutoffs are uniform within each fixed parameter family of the layer theorem.

Proof. Use the retained arbitrary-width chunk-swap primitive for the at most twelve swaps, each of width nG . Its padding, cleanup and all spectator intervals are covered by that interface. For every rotation, apply Lemma 1; its offset arithmetic is polynomial in A . No new record volume is introduced by describing the compact fields. For $\ell_p = \lceil \log_2 p \rceil$, eventually $K \geq G + 4\ell_p + 10$. Then

$$\delta \leq \frac{5}{128p^3}, \quad \frac{A^3}{R} = o(1), \quad \delta A(1 + A/R) = O(p^{-2}).$$

The repair is therefore $O(V)$ per call, including predicate scans and key arithmetic. The elementary highest-bit operation costs $O(V)$. All calls reuse their fixed work tapes and finish in the declared physical field order. $\square$

Application to a fixed binary change of basis. The original phase-frame construction expresses each binary basis change as a fixed number of row additions on m slots of $f = e/m$ axes. Use the two slots of each addition as x, y and the reserved front/back compact fields as controls. The complete basis change and its inverse cost $O(V((e \log p)^\tau + 1))$. The reservations below guarantee their complete ranges, independently of which slots are involved, at every recursive node.

3 A complete recursive layout

Retain the layer input contract: complete address set $[P] \times [2^K]^D \times [S]$, $1 \leq D \leq d$, with polynomial records of $\Theta(rp)$ bits, $d = \Theta(p^\epsilon)$, $K = \lfloor d^\epsilon \rfloor$, $r = 2^{\Theta(p/d)}$, and address/descriptor bit length $O(p)$. All parameter bands are fixed before taking p large. A finite complex network with m selected-coordinate dimensions, W roles and s residual children supplies the same phase decomposition as in the upstream layer construction. Each child uses e/m selected axes and exactly $1/W$ of the parent volume.

Rows and complete compact fields. Let m, W, s denote the complex network constants and keep the layer's global p, d, K at every node. Put

$$G = 4\lceil \log_2 p \rceil + 6, \quad H = dG, \quad q_F = \left\lceil \frac{2H}{K} \right\rceil, \quad q_B = \left\lceil \frac{H}{K} \right\rceil,$$

$$q_0 = \lceil \log_2 W \rceil \lceil \log_m(2d) \rceil, \quad k_0 = \lceil \log_m d \rceil, \quad Q_0 = q_0 + q_F + q_B.$$

If $D \leq Q_0$, apply all D selected kernels individually. Otherwise, process individually the first $q_0 + q_F$ and last q_B selected axes. This uses existing coordinates only. The remaining active axes are a consecutive middle block of length $D - Q_0$.

Regard the first q_0 chunks as the row index. Immediately after it, the next q_F chunks have at least $2H$ bits. Reinterpret these consecutive bits as two H -bit front fields and a spectator remainder; a remainder of length zero is omitted. The last q_B chunks similarly supply one H -bit back field and a spectator remainder. These reinterpretations move no records and preserve the total address space. For a call with f axes per slot, carve $nG = (f - 1)G \leq H$ consecutive bits from each of these fields; collapse their unused bits into spectators. The resulting field list has a fixed number of entries. Both front fields precede all active slots and the back field follows them, so the compact-control proposition applies to every ordered pair of slots. For $f = 1$ no compact fields are used.

The original row range has length $R_{\text{row}} = 2^{q_0 K} \geq W^{k_0}$. For each original prefix, pad this row range to the next multiple of W^{k_0} by adding zero rows. A row includes the entire complete

front fields, active middle block, back field, all spectator coordinates and polynomial record suffix. Padding adds whole such rows, so increases total volume by at most two and does not change any within-row range.

At a node write the row index as $u = Wg + w$, $0 \leq w < W$, and send the entire row to role w . The index u consists solely of the first row region: *no compact-field bit is included in it*. Each role therefore has the same complete within-row address set and exactly $1/W$ of the parent volume. At depth j the remaining row count is divisible by W^{k_0-j} . This invariant survives further splits, padding, and reassembly. Pointwise gates on different roles have aligned g and every remaining coordinate. All compact operations restore the temporary fields before a scalar gate, recursive child, or returned result uses that layout.

The fixed-tape scheduler is unchanged: park only the other role streams at the active stack top, execute the child on a complete role stream, and restore the parked streams on return. Every child has its own current logical volume; copying, parking, rewinds and clearing cost a fixed multiple of that volume. The constant number of compact operations per edge use fixed work areas. Descriptors include the global p, d, K, G, H and the carved field lengths; all have $O(p)$ bits and there are a fixed number of fields. The existing record suffix absorbs polynomial setup at every node and per record.

Each completed invocation acts as the desired C tensor on its active axes and as identity on the reserved and spectator coordinates. It therefore commutes with the earlier individual kernels, including those on bits now interpreted as compact fields. Restoring the row permutation returns the result to each original row separately. A padded zero row is consequently zero at completion and can be deleted exactly as before. Neither temporary field contents nor intermediate scratch payloads were assumed to be zero.

The number of individually processed axes is at most Q_0 , including the fallback $D \leq Q_0$. For large p , $K \geq d^c/2$ and

$$q_F + q_B \leq \frac{3dG}{K} + 2 \leq 6d^{1-c}G + 2.$$

Thus preprocessing, padding and reservation cost

$$O(V(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1)).$$

All reserved axes lie within the original D axes; in particular their total number never exceeds d . This fact also bounds their coefficient dependency depth by the same $8d$ allowance used in the guard proof.

Unrolling the compact-control recurrence. Fix rational $0 < \beta < 1$. Choose $0 < \tau < 1$ and $\max\{0, \log_m(s/W)\} < \sigma < 1$. Stop when $e < d^\beta$ and apply the remaining kernels individually. With $F(e)$ denoting time per current logical volume, the exact row split gives

$$F(e) \leq (s/W)F(e/m) + O(G^\tau e^\tau + 1), \quad F(e) = O(e) \quad (e < d^\beta).$$

Here G and the compact reservations are global parameters, not rebuilt at each node. This is why no factor K^τ appears in the overhead. Repair costs are already included in the node overhead.

Write $b = s/W$. At level j the internal contribution is bounded by $O(G^\tau e^\tau (b/m^\tau)^j)$. Summing through the stopping depth gives

$$O(G^\tau \log(2d)d^\chi), \quad \chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\}.$$

Indeed if $\sigma \leq \tau$ the root bounds every summand; otherwise the last internal level costs at most the root bound times a fixed constant and $d^{(1-\beta)(\sigma-\tau)}$. The additive 1 is absorbed by $G^\tau e^\tau$ at internal nodes. The leaf contribution is at most $O(d^{\sigma+\beta(1-\sigma)})$, by the original volume-weighted argument.

Partition $D - Q_0$ into consecutive powers of m , using its base- m expansion. There are $O(\log(2d))$ pieces. Include their costs and the reservation cost to obtain, per original volume,

$$O\left((\log p)^3[d^\chi + d^{\sigma+\beta(1-\sigma)} + d^{\max\{1-c,0\}} + 1]\right).$$

We used $\log d = \Theta(\log p)$ and $G = O(\log p)$ for each fixed $d = \Theta(p^\epsilon)$ family. Any fixed rational exponents satisfying

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c, 0\} < \lambda'$$

therefore give the complete $O(Vd^{\lambda'})$ layer bound. The strict gaps absorb the logarithms. The small- D fallback obeys the same bound. For fixed rational $\beta = u/v$, implement stopping by the integer test $e^v < d^u$; its polynomial descriptor cost is already included.

3.1 A generalized stopped-depth guard

Use the complex network's own constants m, W, s , independently of the bit network used by swaps. Assume $m \geq 3$ and $2 \leq s < m^5$, as checked below. For rational $0 < \beta < 1$ and $\zeta > 0$, put

$$E = 64(W + m + 1)^3, \quad B = s + E, \quad C_1 = 5 - 4\beta + \zeta, \\ C_0 = \left\lceil \max\{128mB^2, 18mB^2(1 + 1/\zeta)\} \right\rceil, \quad \Delta = \lceil C_0 d^{C_1} \rceil.$$

As in the retained exact-arithmetic guard proof, a dependency-depth upper bound is

$$A(e) \leq sA(e/m) + E \quad \text{internally}, \quad A(e) \leq 8e \quad (e < d^\beta).$$

The fixed node charge E remains additive. All new compact-control, padding and repair operations only permute complete coefficient encodings and introduce no coefficient arithmetic.

For a root $e \leq d$, the number of internal levels is at most $(1 - \beta) \log_m d + 1$. Hence

$$s^j \leq sd^{5(1-\beta)}, \quad A(e) \leq s(8 + E)d^{5-4\beta} \leq 9B^2 d^{5-4\beta}.$$

A root that is already a leaf also satisfies this bound. There are at most $m(1 + 1/\zeta)d^\zeta$ base- m pieces, using $\log d \leq d^\zeta/\zeta$ and $\log m > 1$. Individually processed row, front and back axes are disjoint and number at most d , so their depth is at most $8d$. The outer phases and b^D conversion add at most $\lfloor D/2 \rfloor + 9$ units, including denominator shifts. Consequently

$$A_{\text{layer}} \leq 9mB^2(1 + 1/\zeta)d^{C_1} + 18d \leq C_0 d^{C_1}.$$

The exact integer comparison for the final inequality is included in the certificate. The usual induction gives denominator exponent at most $p + \Delta$ and magnitude at most 2^Δ . Thus $p + \Delta$ fractional bits and $\Delta + 3$ integer/sign bits suffice per component. The condition $\epsilon C_1 < 1$ makes these widths $O(p)$, uniformly over the retained parameter bands. Integer power comparisons implement the rational guard ceilings in polynomial descriptor time.

4 Separating the bit and complex arities

4.1 An independently sized complex network

Keep the paired bit network at $h_b = 50$, with address-swap exponent $\tau = 1 - 296/10^{11}$. Instantiate the original complex construction instead at $h_c = 25$. Its constants, distinguished from those of the bit network, are

$$m_c = 15625, \quad v_c = 2300, \quad N_c = 12167000000,$$

$$W_c = 58645352620000, \quad L_c = 10315500000, \quad s_c = 916333630984500000.$$

Here $W_c = 2v_c^3 + 3v_c^2(v_c z_c + 26)$, $z_c = \binom{22}{3} + 66$, and $L_c = 3v_c^2 \cdot 25 \cdot 26$.

Proposition 2. *The original complex phase interface, including its signed scalar bank exchange, arbitrary-scratch restoration, binary residual bases and endpoint corrections, holds with these constants. Its normalized residual saving supports the strict exponent $\sigma = 1 - 418/10^{12}$.*

Proof. The original construction and residual table apply at this ground size. For an intersection of size j , the center coefficient is $(j - 1)/2$; the side correction is its negative for distinct even-intersection neighbors. Their sum is zero for $j = 0, 1, 2$ and one for $j = 3$. The same eight-step schedule therefore restores arbitrary scratch and gives the elementary bank shear, and the three stages give the signed exchange.

Binary triple indicators have norm one. The orthogonal decompositions in the residual table remain nondegenerate. Every nonzero residual has a norm-one vector: choose a coordinate outside the support of a triple or two neighboring triples (at most six coordinates), or outside an earlier/future tensor indicator (support $3^j < 25^j$); tensor with the remaining norm-one factors. Thus the original nonalternating-space argument supplies the required orthonormal bases. No even-ground-size matching or rational bit-interface positivity is needed for this unchanged complex network.

The rank accounting gives

$$s_c = W_c m_c - 2N_c + 2L_c, \quad \eta_c = \frac{W_c m_c - s_c}{W_c m_c} = \frac{14}{3464399375} > 0.$$

The needed positivity is $L_c < N_c$, not $2L_c < N_c$. Endpoint weight-27 phase corrections and the signed-exchange correction are unchanged. Exact rational logarithm bounds give $\log m_c < 966/100$ and $\eta_c > (418/10^{12})(966/100)$. The inequality $e^{-x} > 1 - x$ then yields $s_c/W_c < m_c^\sigma$. The certificate also checks $2 \leq s_c < m_c^5$. $\square$

The bit and complex dimensions need not agree. The bit interface supplies swaps at arbitrary binary widths. The complex recursion uses only that interface's exponent τ and fixed tape guarantee, while its own branching, role split and depth use m_c, W_c, s_c . All occurrences of m, W, s within the revised layer proof denote these complex constants. The retained assembly consumes their time exponents and completed-layer contract, so it introduces no equality requirement between the two arities.

5 Parameters and complete assembly

Put $a = 296/10^{11}$ and $a_c = 418/10^{12}$. Choose

$$\begin{aligned} \tau = 1 - a, \quad \sigma = 1 - a_c, \quad \epsilon = \frac{1999}{10000}, \quad c = \frac{1}{5}, \quad \beta = \frac{1}{1000}, \quad \zeta = \frac{1}{10000}, \quad \delta = \frac{1}{10^6}, \\ C_1 = \frac{49961}{10000}, \quad \lambda = 1 - \frac{1671}{4 \cdot 10^{12}}, \quad \lambda' = 1 - \frac{167}{4 \cdot 10^{11}}, \quad \kappa = \frac{83}{10^{12}}. \end{aligned}$$

The layer exponents are

$$\chi = \tau + (1 - \beta)(\sigma - \tau), \quad \chi_{\text{leaf}} = \sigma + \beta(1 - \sigma), \quad \chi_{\text{reserve}} = 1 - c = \frac{4}{5}.$$

Exact rational comparisons show $\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1$ and $\max\{\chi_{\text{leaf}}, \chi_{\text{reserve}}\} < \lambda'$. Also $\epsilon C_1 = 99872039/10^8 < 1$. This supplies the same completed normalized-butterfly contract as

the original layer theorem, with exact evaluation before truncation and the same disk and error guarantees.

The retained Gaussian choice satisfies $\alpha^4 \theta_i > 8b > p = 6b$. It gives

$$\alpha = \Theta(p^{11999/40000}), \quad \gamma = O(p^{15997/20000}) = o(p).$$

Since $\epsilon < 1/5$, $b \geq 2^{40}$ implies $46b^{(1+3\epsilon)/2} \leq b/4$: indeed $184 < 2^8$. Prime selection still has $1 - 2\epsilon = 3001/5000 > 0$. The polynomial record length remains superpolynomial, while $\epsilon(1+c) < 1$ ensures $K = o(\ell)$ and the required address-size bounds. The new reserved-axis count is $O(\log d + d^{1-c} \log p) = o(d)$. For small D the individual-kernel fallback was already charged.

The seven normalized assembly margins are unchanged in form:

$$\begin{aligned} g_1 &= 1 - \epsilon(1 + c), & g_2 &= \epsilon c(1 - \tau), \\ g_3 &= \epsilon(1 - \lambda'), & g_4 &= (1 - \tau)(1 - \epsilon), \\ g_5 &= 1/4 - \delta - 5\epsilon/4, & g_6 &= 1 - \delta - \epsilon, \quad g_7 = \epsilon. \end{aligned}$$

All reservation, compact movement and repair costs have been included in the layer bound, so require no extra assembly term. Exact arithmetic gives

$$G_* := \min_i g_i = g_3 = \frac{333833}{4 \cdot 10^{15}}, \quad G_* - \kappa = \frac{1833}{4 \cdot 10^{15}} > 0.$$

This strict gap absorbs all remaining fixed powers of $\log p$. Hence the retained multiplication assembly yields

$$T(n) = O(n(\log n)^{1-83/10^{12}}), \quad 2^{-34} < \frac{83}{10^{12}} < 2^{-33},$$

conditional on the identified upstream and extension proofs. The simpler dyadic statement with $\kappa = 2^{-34}$ follows immediately.

Exceptional streams must be charged locally. The original manuscript separately bounded the global sorting work using an exponentially small 2^{-K} bad fraction. The compact gadget instead has an $O(p^{-3})$ fraction. We do not reuse that global $o(V)$ claim. For each call, key sorting costs $O(V_{\text{call}} p^{-2})$ and extraction and reinsertion cost $O(V_{\text{call}})$. Those costs were included in every node of the volume-weighted recurrence. Summing that recurrence, including all children and base- m pieces, already proves the stated layer cost. Polynomial setup per record is independently absorbed by the superpolynomial record suffix, as in the retained proof. Thus there is no uncharged factor equal to the number of recursion nodes.

Earlier targets and the next ceiling. The exact checker also verifies the previously hypothetical 2^{-39} parameters using the original $h = 50$ complex saving, and the 2^{-34} parameters using $h = 25$, now with the compact cost and reservation term. For the fixed $h = 25$ complex motif, its actual saving is at most an exact enclosure of approximately $4.1847990372 \cdot 10^{-10}$. The retained Gaussian margin forces $\epsilon < 1/5$, while the leaf margin forces $1 - \lambda' < (1 - \beta)(1 - \sigma) < 1 - \sigma$. Consequently this motif and these assembly inequalities have the strict ceiling

$$\kappa < \frac{a_c^{\text{actual}}}{5} < 8.369598075 \cdot 10^{-11} < 2^{-33}.$$

The displayed witness exceeds 99% of this upper enclosure. This is not an optimality theorem for other complex networks, stronger leaf routines, or different multiplication algorithms. A next dyadic improvement within this strategy requires improving one of those ingredients.

6 Reproducibility and review boundary

The principal artifacts are

- `scripts/audit_compact_controls.py`: modular address permutation, inverse, control order and exceptional repair;
- `scripts/compact_control_layer.py`, with certificate `certificates/compact-control-layer.json`: reservations, recurrence, guard, motif exponents, seven margins and the scoped ceiling;
- `patches/compact-control-34.patch`: an independent patch against the pinned source, preserving the old wide-slot lemma for its appendix users;
- the compact-control test suites, including explicit whole-row padding and recursive splitting, two-piece stream rotations, all exponent orderings, dirty temporaries and bad-address repair.

The patch updates the layer’s own complex constants, the generalized guard, reserved-coordinate accounting, numerical setup, and global exceptional-stream paragraph. The published 2^{-59} artifacts and pinned upstream files remain unchanged. Neither the tests nor the PDF build constitute independent proof review or a formalization of the upstream theorem.